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Metrological Evaluation of Software-Defined Radios (Adalm-Pluto and LimeSDR usb) in Radio Frequency Signal Generation

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Abstract

In the automotive industry, products must be tested and verified before being launched; currently, products such as automotive radio are usually tested with expensive metrology equipment, such as the Rohde & Schwarz SFE100, which produces radio frequency signals. This study assessed the feasibility of replacing the SFE100 with two inexpensive devices, Adalm-Pluto and LimeSDR usb, which are software-defined radios. For this evaluation, a study was carried out to determine specific metrological characteristics and then to compare with those of metrological equipment. Based on the results and tests of the study, we conclude that SDRs can generate test signals (FM, DAB, DVB, ISDB and SXM) with uncertainties of 0.5dBm and a spectral density ratio comparable to SFE100, thereby enabling their replacement. Furthermore, the evaluation datasets of all equipment have been publicly available [1], enabling the scientific community to extract the metrological uncertainties for all signal carriers generated by the device. The result of this study proposes a methodology for testing auto radios that is innovative and can be applied to a production line based on the principles of Industry 4.0.

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Keywords: RF Signals; Software-Defined Radio (SDR); Spectrum Analysis; Industrial Testing; Metrological Devices.

1. Introduction

Nowadays infotainment systems have several functions, one of which is the tuner. Furthermore, these functions require production quality control, in which case several radio frequency (RF) signals (AM, FM, DAB, DVB, ISDB, and SXM) must be injected into the device under test (DUT). This verification process uses expensive metrological equipment such as the Rohde & Schwarz SFE100 [2]. The aim of this article is to evaluate the replacement of

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these expensive devices with cheaper equipment such as software-defined radios (SDRs), which can also produce quality signals within the range required for these tests. These SDRs lack certified metrological characteristics and are therefore unable to be deployed in the industrial environment for the production of RF signals. Therefore, it is necessary to conduct an extensive metrological evaluation of these devices. In addition, each of them can cover a frequency and amplitude range mapping. Using this approach, we aim to assess the Type A uncertainty of each output and test whether these devices can replace the SFE100 in all tests, and demonstrate their feasibility scientifically.

Initial evaluation settings were implemented enabling the signal injection devices (SDR or SFE100) to be connected to spectrum analyzers and test management devices (author software computers). Furthermore, the whole spectrum and amplitude were injected through iterative mapping and each iteration recorded a Fast Fourier Transformation (FFT) from a spectrum analyzer. With each device mapping, a dataset is created and made public [1], and a metrological evaluation is carried out. Moreover, the three metrological characteristics (amplitude (dBm), frequency (Hz), carrier-to-noise spectral density ratio (CN0) and uncertainty values) were extracted for the entire range of devices. The main contributions of the paper are as follows:

- Metrological mapping dataset [1] for SFE100 [2], LimeSDR [3], and PlutoSDR [4];
- Metrological evaluation of LimeSDR and PlutoSDR for the replacement of SFE100 in production grade test signals.

The remainder of the paper is organized as follows. Section 2 presents the state-of-the-art application of SDRs in different aspects of use. The dataset creation and model implementation is described in section 3. Experiments are described in Section 4, with their corresponding results. The discussion is presented in Section 5. Section 6 concludes the paper and presents some future work.

2. Related Work

Molla et al. [5] drew up the concept of "Software Radio" to describe the transition from a radio system based on the design of hardware to a system where software plays an important role in defining most functions. The main advantage of software radio is its flexibility, allowing users to create radio waveforms and modify them by simply changing software (SW) without changing the underlying SDR platform, including the hardware (HW) and operating environment required for the operation of waveform applications [6][7].
The ultimate goal is, although a bit difficult, to achieve the ability to communicate at any desired frequency, bandwidth, modulation and data rate by loading the appropriate SW [8]. The rapid development of communication standards is one of the main driving factors of business communication. Consequently, upgrading the SW of the base station is an attractive option, compared with the costly replacement of the HW. There are many reasons for adopting SDR solutions [9]. In the military field, the life of communication systems is longer than that of commercial systems. SDR also proves to be useful for space applications and allows flexibility to update the SW of deployed satellite communications equipment to align with advances in communication algorithms and standards [10][11][12]. This facilitates changes in communication functions and allows multiple applications throughout the lifetime of the satellite operation. Chiper et al. [13] provides a comprehensive review of existing literature on drone detection and defense systems.
In this overview, various methods are included such as RF, acoustic, optical and radar. The present paper focuses particularly on RF-based systems implemented on the SDR platform. In a company, the implementation of the measurement system is often driven by the need to ensure satisfactory quality in the production process and use a measurement tool for inspection. It has been shown in many studies that metrology is a value-added activity in all stages of production. However, metrology has been sometimes perceived as unproductive and manufacturing engineers often seek to minimize the costs associated with metrology.
This paper explores the role of metrology in production and emphasizes its potential value creation. It presents various approaches for assessing the advantages of metrology and provides instructions to metrologists to effectively promote metrology using economic arguments [14]. The concept of metrology in modern times has been widely accepted and applied. Metrology is defined as a scientific field that encompasses both theoretical and practical aspects of measurement, including a wide range of areas and the uncertainty of measuring associated with it [15][16]. It provides a

comprehensive framework for understanding and implementing accurate and reliable measurement practices in various areas of BIPM application [17].
In all countries, national measurement systems (NMSs) are networked with laboratories, calibration facilities, and accreditation bodies, responsible for the implementation and maintenance of metrological infrastructure in the country [18]. NMS has an important role in the formation of national measurement practices and in the establishment of recognition with in the international community. This in turn has a significant impact on various aspects of society including economy, energy, environment, health, manufacturing, industry and consumer confidence [19]. The impact of metrology on trade and economy is especially remarkable, as a set of agreed measurement systems is essential to facilitate fair trade [20].

3. Implementation

In this section, we will present our implemented methodology. Covering signal generation in the studied devices (i.e., SFE100, PlutoSDR, and LimeSDR), detailing, respectively, how the communication with the devices was made, signals acquisition, and evaluation metrics. Figure 1 a) summarizes the entire development pipeline. Figure 1 b) shows the flowgraph developed in GNU Radio.

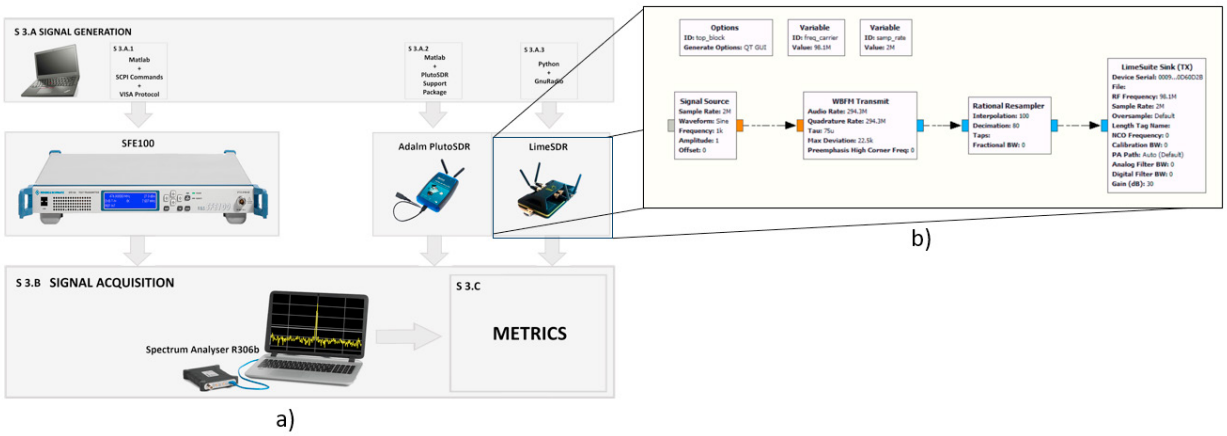


Fig. 1. a) Development pipeline is presented. b) Block developed in the GNU Radio software for communication with LimeSDR.

3.1. Signal Generation

3.1.1. SFE100

In order to characterize this equipment to serve as a basis for the SDR studies, a mapping was carried out, consisting of the injection of carrier waves into the entire frequency and amplitude range, with the level of resolution allowed by the equipment (as shown in Table 1). SFE100 control was established with an internally developed Matlab script, using SCPI commands (i.e., standard commands for programmable instruments) through the VISA protocol.

Table 1. Specifications guaranteed by the manufacturer for the equipment's used in this study [2][3][4].

	Frequency Range	Amplitude Maximum level	Bandwidth
PlutoSDR	70MHz - 6GHz	7dbm	56MHz
LimeSDR	100kHz - 3.8GHz	10dbm	61.44MHz
SFE100	100kHz - 2.7GHz	27dbm	35MHz

3.1.2. PlutoSDR

Characterization followed the same principle as in SFE100 (as shown in Table 1). Communication with the equipment was carried out using an internally developed MATLAB script, which has used communication packages that support communication with PlutoSDR (Support Package PlutoSDR [21]).

3.1.3. LimeSDR

Similar to SFE100 and PlutoSDR, characterization was defined by Table 1. Moreover, LimeSDR communication was made through an internally developed Python script, using GNU Radio Software [22]. In Figure 1 b) shows the flowgraph developed in GNU Radio that produces the different signals, since GNU Radio has the LimeSuite Sink (TX) block dedicated to communication with LimeSDR. The way to control the gain of the signals is through the LimeSuite Sink (TX) block, which accepts gain values from 0 to 60 dB.

To make the conversion from the GNU Radio blocks flowgraph to a Python script, we followed the processing available on the GNU Radio Wiki Web page [23]. Moreover, the Python script is generated automatically once the RUN button on the GNU Radio application is activated. The following example is presented, already completed in Python format:

```
#!/usr/bin/env Python3
from gnuradio import gr, audio, analog
class my_top_block(gr.top_block):
    def __init__(self):
        gr.top_block.__init__(self, "Not titled yet")
        samp_rate = 2e6
        db = 30
        amp_signal = 1
        freq_signal = 1e3
        freq_carrier = 98.1e6
        self.samp_rate = samp_rate
        self.rational_resampler_xxx_0 = filter.rational_resampler_ccc(
                interpolation=100,
                decimation=80,
                taps=None,
                fractional_bw=None)
        self.limesdr_sink_0 = limesdr.sink('0009072C00D60D2B', 0, '', '')
        self.limesdr_sink_0.set_sample_rate(samp_rate)
        self.limesdr_sink_0.set_center_freq(freq_carrier, 0)
        self.limesdr_sink_0.set_bandwidth(5e6, 0)
        self.limesdr_sink_0.set_digital_filter(samp_rate, 0)
        self.limesdr_sink_0.set_gain(int(db), 0)
        self.limesdr_sink_0.set_antenna(255, 0)
        self.limesdr_sink_0.calibrate(2.5e6, 0)
        self.analog_wfm_tx_0 = analog.wfm_tx(
            audio_rate=294.3e6,
            quad_rate=294.3e6,
            tau=75e-6,
            max_dev=22.5e3,
            fh=0,
        )
        self.analog_sig_source_x_0 = analog.sig_source_f(samp_rate, analog.GR_SIN_WAVE,
            ↪ freq_signal, amp_signal, 0, 0)
        self.connect((self.analog_sig_source_x_0, 0), (self.analog_wfm_tx_0, 0))
        self.connect((self.analog_wfm_tx_0, 0), (self.rational_resampler_xxx_0, 0))
        self.connect((self.rational_resampler_xxx_0, 0), (self.limesdr_sink_0, 0))
if __name__ == '__main__':
    try:
        my_top_block().run()
    except [[KeyboardInterrupt]]:
        pass
```

After importing the GNU Radio modules, a derived class (my_top_block) is defined from the ”gr.top_block” class, which is called the ”__main__” function. In a first stage, signal related variables, such as sample rate, ”samp_rate”,

amplitude, "amp_signal", frequency, "freq_signal", and carrier frequency, "freq_carrier" are initialized. Moreover, the Signal Source function, "self.analog_sig_source_x_0" is created with these parameters, as well as the WBFM Transmit function, "self.analog_wfm_tx_0", and the Rational Resampler function, "self.rational_resampler_xxx_0" with their respective parameters. Finally, "self.limesdr_sink_0" function is the LimeSuite Sink (TX) block with some extra parameters to transmit the signal, which will be generated through the "self.connect" function.

3.2. Signal Acquisition

Each one of the equipment's was connected to the Spectrum Analyzer R306B [24] through a 15cm RG-400U cable, whose maximum typical attenuation, depending on the cable manufacturer, is 0.1dB at 2.4GHz [25], thus not affecting the measurements performed. Furthermore, the signals were collected using the spectrum analyzer, which provides the function of saving each of the representative FFTs of those signals in an csv file (that is, equation 1). The FFTs are composed of a total of 800 bins acquired with 100Hz of resolution bandwidth (RBW) in order to distinguish frequency variations of this order of magnitude, as well as having a low noise threshold, which allows capturing the signal even for low values of signal amplitude. As mentioned above, for experiments, an FFT is acquired for each generated signal, from the minimum limit achieved by each equipment (d_i and h_i), and the maximum limit (d_j and h_j), respectively. Here, d and h represent the frequency indexing and amplitude, respectively.

$$\begin{aligned} FFT_{d,h} &= \{dBm_{1:801}, hz_{1:801}\} \\ d &= d_i, \ldots, d_j \\ h &= h_i, \ldots, h_j \end{aligned} \tag{1}$$

3.3. Metrics

The metrics used in this study to evaluate the signals previously presented are: uncertainty in frequency (μHz and σHz), uncertainty in amplitude (μdBm and σdBm), and ratio between the carrier power (C) and the spectral noise power density (NO).

3.3.1. Uncertainty

Uncertainty can be calculated taking into account two types of components: types A and B. Type A uncertainty is related to the reliability of the measurement measured through the analysis of statistical fluctuations, and this uncertainty is related to the concept of standard deviation, which measures how much the data obtained varies in relation to the average of the data itself (this proves that the uncertainty does not depend on the real value). Type B uncertainties depend on other factors in addition to randomness, such as maturity, intuition, and the knowledge the user has accumulated about his instruments and measurement procedures. In this study, only Type A uncertainty will be considered. Consequently, the number of repetitions of a measurement is reduced to a number ranging between 3 and 10, N, where it is necessary to apply the expanding factor k, present in the table t-Student [26], which takes into account the required level of confidence and the number of effective degrees of freedom of the distribution (N-1). Moreover, we provide a statistical measurement as shown in equation 2.

$$Y = \mu \pm (\sigma \times k) \tag{2}$$

Uncertainty was calculated for Frequency (Hz) and Amplitude (dBm), for each μ and σ were extracted for each of the generated signals, in each of the tested equipment, as shown in equations 3 and 4, where d and h represent the indexing of generated signal amplitude and frequency, respectively.

$$a = \max\left(FFT_{d,h}\{dBM\}\right)$$
$$\mu dBm_{d,h} = \frac{\sum_{n=1}^{N} a}{N} \quad (3)$$
$$\sigma dBm_{d,h} = \sqrt{\frac{\sum_{n=1}^{N}\left(a - \mu dBm_{d,h}\right)^2}{N}}$$

$$f = \operatorname{argmax}\left(FFT_{d,h}\{dBM\}\right)$$
$$\mu Hz_{d,h} = \frac{\sum_{n=1}^{N} FFT_{d,h}\left\{hz_f\right\}}{N} \quad (4)$$
$$\sigma Hz_{d,h} = \sqrt{\frac{\sum_{n=1}^{N}\left(FFT_{d,h}\left\{hz_f\right\} - \mu Hz_{d,h}\right)^2}{N}}$$

3.3.2. CN0

This parameter evaluates the quality of the signal, indicating how much higher the signal power (that is, in this case the carrier) is compared to the noise. CN0, equation 5, corresponds to the ratio between C (carrier power) and N0 (spectral noise power density), that is, the noise power normalized by bandwidth.

$$C = \max\left(FFT_{d,h}\{dBM\}\right)$$
$$f = \operatorname{argmax}\left(FFT_{d,h}\{dBM\}\right)$$
$$N0 = \frac{\left(\frac{\left(\sum_{n=1}^{f-1} FFT_{d,h}\{dBM\}\right)+\left(\sum_{n=f+1}^{800} FFT_{d,h}\{dBM\}\right)}{800-2}\right)}{RBW} \quad (5)$$
$$CN0_{d,h} = 10\log_{10}\left(\frac{C}{N0}\right)$$

4. Experiments and Results

4.1. Data Generation

Uncertainty metrics were focused in Type A, thus it was required to collect 5 measurements, N, for each generated signal (that is, 5 degrees of freedom). Furthermore, the FFTs collection process was repeated 5 times to obtain the uncertainties (i.e. equations 3, 4 and 5) as metrics for evaluating frequency, amplitude, and CN0, later expanded, k, to a confidence level of 95%. Furthermore, according to the t-student table with 5 degrees of freedom for a 95% certainty, the multiplier value is 2.015048. The FFTs were recorded on gain and frequency requirements (Table 2). The frequency was tested in each instrument with logarithmic increases from the minimum limit (h_i) to the maximum (h_j), and the amplitude with increases of 2 from the minimum limit (d_i) to the maximum (d_j), for each instrument, respectively.

Table 2. The datasets collected for the 3 devices followed the maximum and minimum frequency and gain limits according to the capabilities of each device. Minimum gain limit (d_i); maximum gain limit (d_j); lower frequency limit (h_i); maximum frequency limit (h_j).

	d_i	d_j	h_i	h_j
SFE100	-110 dBm	0 dBm	100000Hz	2685643818Hz
PlutoSDR	-89 dB	-1 dB	70357295Hz	5987169134Hz
LimeSDR	1	57	30024617Hz	3746050032Hz

One way to evaluate the quality of the signals is by observing the FFTs of the carrier waves generated by the 3 devices and comparing them (Figure 3 a)), these 3 FFTs were produced by the 3 devices under the same conditions, for a frequency of 98MHz and a gain of -22dBm.

The generated FFTs resulted in a number of 11611, 10151 and 23561 FFTs for PlutoSDR, LimeSDR and SFE100 respectively. From the FFTs information, it was also possible to calculate the uncertainty at the level of frequency, amplitudes, and CN0. Thus, we generate the MoLa SDRs Calibration Dataset (that is, made publicly available [1]).

4.2. Uncertainty

As mentioned previously, μ and σ were calculated for the amplitude, frequency and CN0 and then σ was expanded to a confidence level of 95%. Figures 2 a), 2 b), and 2 c) present the individual, and overlapped, uncertainty maps for each device. Moreover, a complete uncertainty map for all devices and metrics is shown in Figure 3 b).

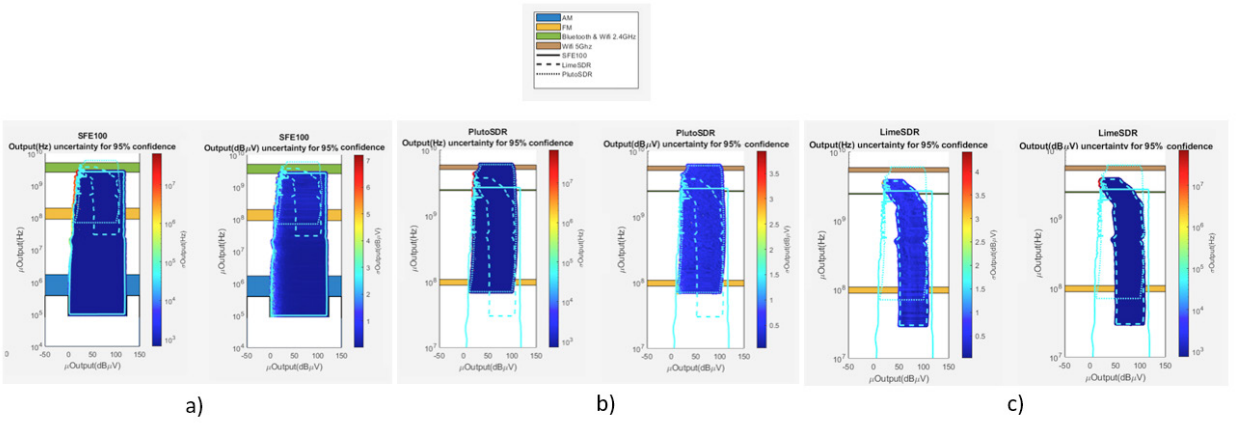


Fig. 2. a) SFE100. Uncertainty plots at frequency (Hz) and amplitude (dBuV) level with 95% confidence level. b) PlutoSDR. Plots of the uncertainties at frequency (Hz) and amplitude (dBuV) level with 95% confidence level. c) LimeSDR. Plots of the uncertainties at frequency (Hz) and amplitude (dBuV) level with 95% confidence level.

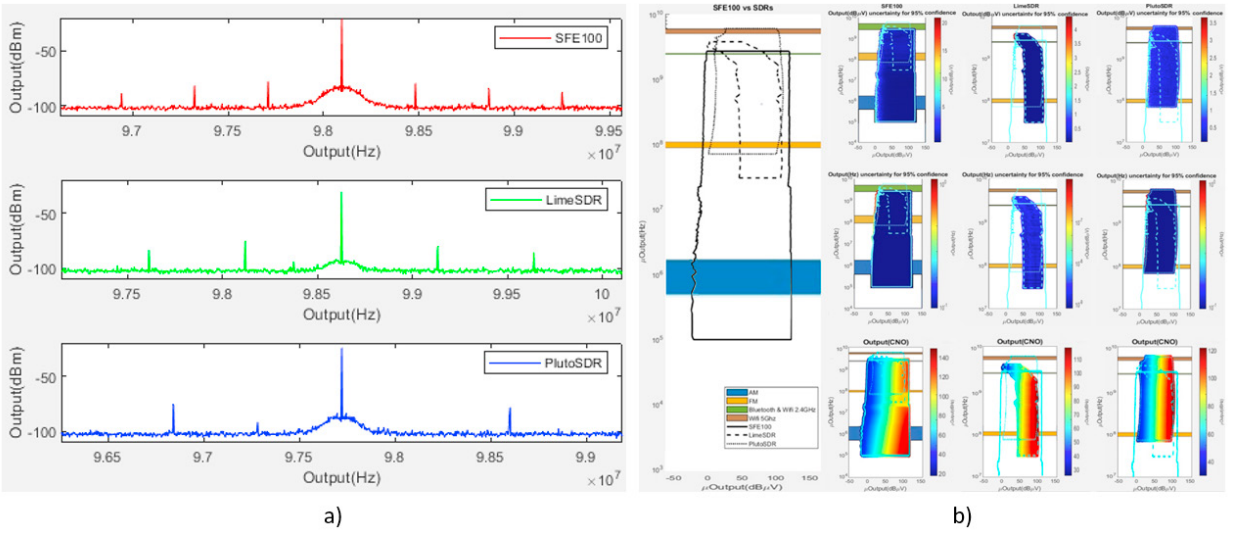


Fig. 3. a) FFTs for the 3 devices (SFE100, LimeSDR and PlutoSDR) for a carrier wave of -22dBm gain produced at a frequency of 98MHz. b) Summary graph of the potential of all the equipment in the production of unmodulated signals.

4.3. Use-Cases

Despite the thorough metrological mapping of the 3 devices, additional use case evaluations were performed. Focusing on specific commercially available frequency bands and the behavior of each device through σdBm and $\mu CN0$ metrics: AM at 1kHz: Figure 4 a); FM at 98MHz: Figure 4 b); DAB at 230MHz: Figure 5 a); DVB at 174MHz: Figure 5 b); ISDB at 470MHz: Figure 6 a); GSM at 850MHz: Figure 6 b); SXM at 2.34GHz: Figure 7.

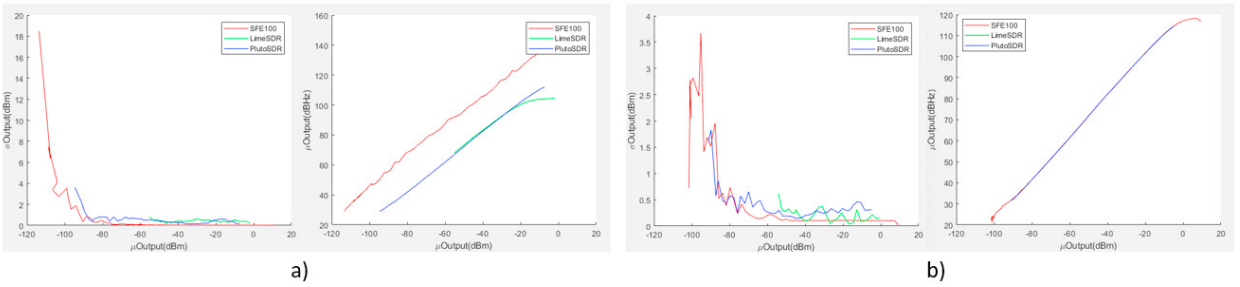


Fig. 4. a) AM signal σdBm and $\mu CN0$ metrics for the 3 devices at 1kHz, from left to right respectively. b) FM signal σdBm and $\mu CN0$ metrics for the 3 devices at 98MHz, from left to right respectively.

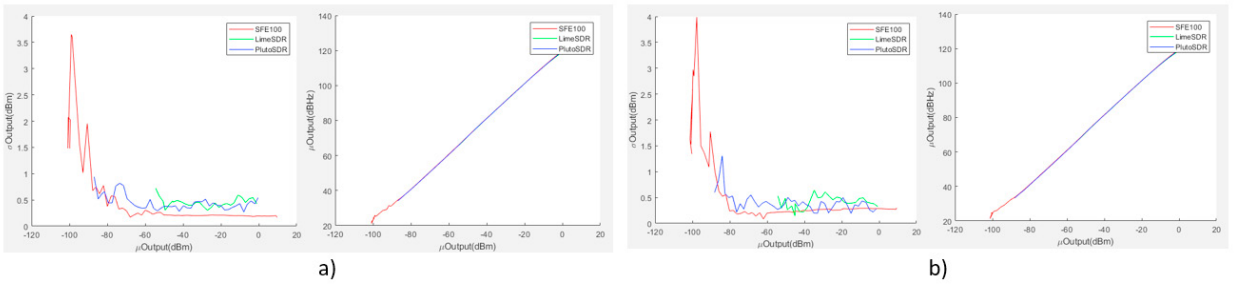


Fig. 5. a) DAB signal σdBm and $\mu CN0$ metrics for the 3 devices at 230MHz, from left to right respectively. b) DVB signal σdBm and $\mu CN0$ metrics for the 3 devices at 174MHz, from left to right respectively.

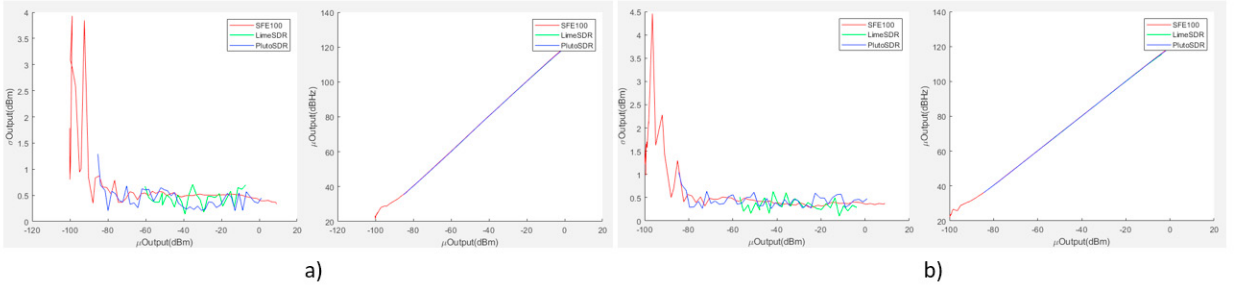


Fig. 6. a) ISDB signal σdBm and $\mu CN0$ metrics for the 3 devices at 470MHz, from left to right respectively. b) GSM signal σdBm and $\mu CN0$ metrics for the 3 devices at 850MHz, from left to right respectively.

5. Discussion

Section 3 showed our implemented methodology, from the generation of unmodulated signals, more precisely carrier signal, in the three devices under study (i.e. SFE100, PlutoSDR, and LimeSDR), to its acquisition and metrics

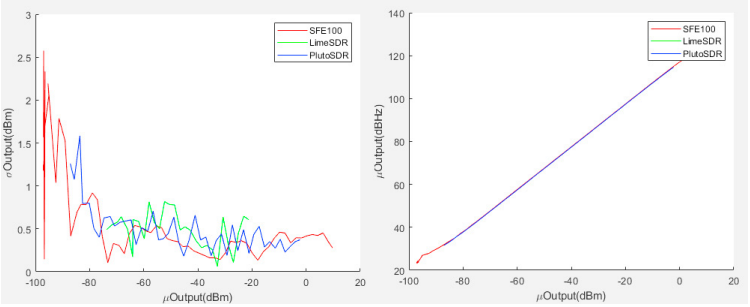


Fig. 7. SXM signal σdBm and $\mu CN0$ metrics for the 3 devices at 2.34GHz, from left to right respectively.

estimation. These signals were generated in the entire frequency and amplitude range indicated by the manufacturer of each device (see Table 1).

The presented methodology used internally developed signal generation scripts, using the most convenient communication protocol for each equipment. The communication and transmission of signals with the SFE100 was made through a MATLAB script using SCPI commands through the VISA protocol. With PlutoSDR communication and transmission was made with MATLAB using a specific package [27] for communication with this SDR. Finally, the communication with LimeSDR was done through a Python script that was achieved through the GNU Radio software, which is a specific software for communication with certain SDRs. The signals on the three devices were acquired by Spectrum Analyzer R306B, where the FFTs of each generated signal were saved and stored for each device. In order to analyze these FFTs we used specific metrics to analyze and compare the quality of the signals produced, these metrics are the uncertainty in frequency, amplitude and CN0.

Section 4 presented the metrological experiments and results, related to metrics throughout the device range and specific commercial frequency bands. In a first qualitative comparison of each equipment, we can look at Figures 2 a), 2 b), 2 c) and 3.

Regarding σdBm all devices present a uniform uncertainty throughout the range, with values lower than 0.5dBm. The same is visible for σHz, with values lower than $10^{e^{-1}}$Hz, for the 3 devices. $\mu CN0$ is higher in SFE100, thus better, however all devices present a linearly proportional CN0 to dBm. SFE100 and PlutoSDR reach its entire manufacturer amplitude/frequency range in a very linear configuration, however LimeSDR lacks this linear behaviour for dBm for frequencies above 2GHz, showing a loss of amplitude as frequency increases. Looking into the commercial frequency bands, SFE100 shows better σdBm across the entire amplitude and frequency bands (i.e. less oscillation and/or lower value). Regarding $\mu CN0$ all devices are very identical, except in AM where SFE100 shows better, i.e. higher, values.

6. Conclusions and Future Work

This paper presents an evaluation approach of SDRs (i.e. PlutoSDR and LimeSDR) in the generation of unmodulated signals, with this evaluation we intend to compare and understand the viability/possibility of replacing the SFE100 metrological equipment, an expensive equipment, in the production line of car radio tests.

A metrological evaluation methodology was implemented for the SDRs, for each a deep signal generation mapping was performed across its entire amplitude and frequency range. In addition, FFTs for each signal were recorded to estimate specific Type A uncertainty metrics. The entire metrological mapping dataset was made publicly available [1], to allow the scientific community to associate each generated signal with its real Type A uncertainty or propose advanced calibration techniques based on artificial intelligence to mitigate the lack of linear behavior. Finally, a qualitative and quantitative comparison was made between the 3 devices, regarding its manufacturer specifications and commercial frequency bands. We conclude that PlutoSDR outperforms LimeSDR in most cases; however, LimeSDR also manages to perform satisfactorily in most situations. In the industrial production environment, a high level of robustness is required of the test equipment in the sense that it must be able to work uninterrupted for extended periods, while keeping the performance within the specifications. Thus, as a future work to consolidate the replacement of expensive metrological equipment by SDRs, it would consist in verifying the characteristics of the signal generated by

the SDR during several days uninterruptedly, for different values of frequency and amplitude. On the other hand, because this type of SDRs typically does not comply with all supplier specifications, it becomes important to understand whether they resemble the same samples with the same HW. Thus, it would be possible to infer the reproducibility of this type of equipment.
Based on the developed work, new SDR-based testing approaches were explored, verifying that for non-signalling tests it is possible to replace metrological equipment, thus allowing a drastic reduction in the total cost of testing. SDRs can generate production test signals (FM, DAB, DVB, ISDB and SXM), with an uncertainty of 0.5dBm, for a confidence level of 95%, and a CN0 comparable to the SFE100, thus allowing its replacement.

Acknowledgments

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